Final Report: System Dynamics Modeling of the Effect of Greenhouse Gases on Himalayan Ice Cover in Nepal

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1 Summary

Glaciers in the Himalayas of Nepal are melting faster over time. This is mainly because of rising greenhouse gases, pollution like black carbon from nearby regions, and changes in how much sunlight the ice reflects. This report builds a system dynamics model to understand how different factors like carbon in the air, glacier size, glacial lakes, forest cover, and black carbon interact from 1980 to 2080.

The model shows four main processes that keep speeding up glacier melting. First, when ice melts, less sunlight is reflected and more heat is absorbed, which causes even more melting. Second, as glacial lakes grow, they erode the glacier edges and increase melting further. Third, rising temperatures can release carbon trapped in frozen soil, which adds more greenhouse gases to the atmosphere. Fourth, black carbon settles on ice, making it darker and causing it to absorb more heat and melt faster.

There are also two natural processes that slow things down a bit. Oceans absorb some carbon from the atmosphere, and forests also help by storing carbon. However, these are not strong enough to stop the overall trend.

If current trends continue, glacier volume is expected to drop from 312 km^3 in 1980 to about 155 km^3 by 2080, which is nearly a 50 percent loss. Reducing black carbon emissions turned out to be one of the most effective actions, as it lowered the risk of dangerous glacial lake outburst floods and helped preserve more ice. The best results came when black carbon reduction was combined with slowing down overall human emissions.

2 Problem Definition and Reference Mode

2.1 The Problem

Nepal has more than 3,800 glaciers spread over a large area, storing a huge amount of ice that provides water to millions of people in the Himalayan region [5]. However, rising greenhouse gases are increasing global temperatures, and these glaciers are very sensitive to such changes.

In this model, we study how different parts of the system are connected. These include carbon in the atmosphere, temperature in glacier regions, glacier size, glacial lakes, forests, and black carbon pollution. The model focuses on Nepal, but the main driver is global emissions, which Nepal cannot directly control. The aim is to understand how these global changes affect glacier loss and the risk of glacial lake outburst floods over time.

2.2 Reference Mode

Figure 1 shows how glacier volume changes in the base case scenario. The decline is slow during the 1980s and 1990s, but after 2000 it becomes much faster as temperatures continue to rise. This creates an S shaped pattern where loss starts gradually and then speeds up over time. This trend is similar to what has been observed in real data, where glacier melting has increased significantly in recent decades [8, 5].

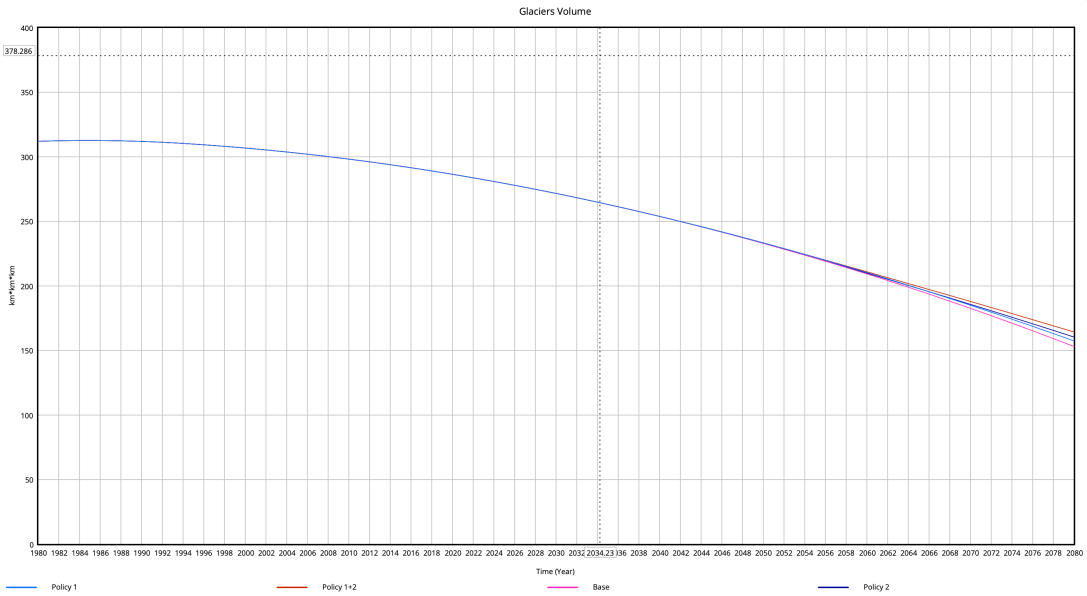

Figure 1: Reference mode: Glaciers Volume declining from 312 km^3 in 1980 to approximately 155 km^3 by 2080 under base case conditions.

2.3 Dynamic Hypothesis

We assume that glacier loss is not just a direct result of rising emissions, but is driven by feedback processes that make the problem worse over time. When glaciers melt, darker surfaces like rock and water are exposed, which absorb more heat and increase temperature further. Growing glacial lakes also add to the problem by warming and eroding the glacier edges. In addition, once temperature rises beyond a certain level, frozen soil starts releasing carbon dioxide, which increases greenhouse gases even more. Because of these effects, glacier loss becomes faster over time even if emissions grow steadily.

3 Qualitative Mapping

3.1 Causal Loop Diagram

Figure 2 shows the causal loop diagram developed at the start of the project. It maps the conceptual causal structure of the system before any equations were assigned, identifying the key variables and the direction of influence between them.

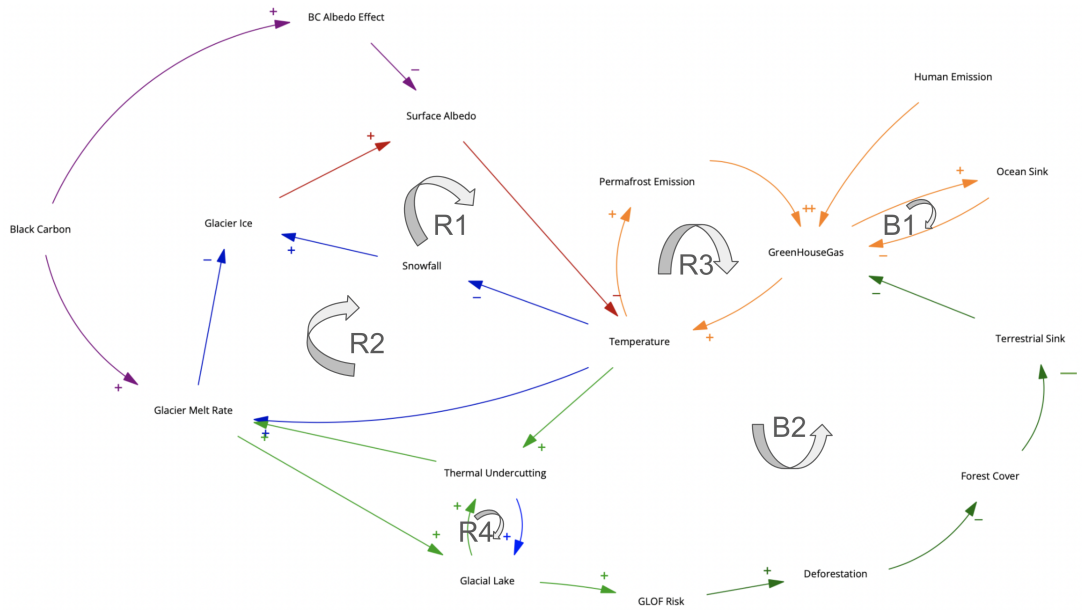

Figure 2: Causal Loop Diagram of the Nepal glacier system

3.2 Feedback Loop Analysis

R1: Ice Albedo Reinforcing Loop

This is the main loop that drives glacier melting in the system. When greenhouse gas levels increase, temperature rises. Higher temperature leads to faster glacier melting, which reduces both glacier volume and snow cover. As the total amount of ice decreases, the surface becomes less reflective. This means less sunlight is reflected back and more heat is absorbed. As a result, temperature increases further, which causes even more melting. This cycle keeps repeating and makes the process faster over time. Because of this loop, glacier loss does not happen at a steady rate. Instead, it accelerates, which is why the curve becomes steeper over time. Without this loop, glacier loss would have followed a more direct and slower pattern.

R2: GLOF Undercutting Reinforcing Loop

As temperature increases, glacier melting also increases. Around 30 percent of this meltwater flows into nearby glacial lakes instead of rivers. This causes the lakes to grow larger. A larger lake holds more warm water, which increases the erosion at the glacier edge. This process, known as thermal undercutting, causes more ice to break off into the lake. As more ice enters the lake, the lake grows even larger. This creates a cycle where lake growth leads to more melting, and more melting leads to faster lake growth. This process becomes dangerous because it increases the risk of glacial lake outburst floods. The risk grows slowly at first, but after a certain point, it increases very quickly.

R3: Permafrost Carbon Trap Reinforcing Loop

This loop becomes active only after temperature rises above a certain level. Once temperature crosses about 1.5 degrees Celsius above the baseline, frozen soil in the region begins to thaw. This soil contains stored carbon, which is released into the atmosphere as carbon dioxide. This increases greenhouse gas levels, which raises temperature further. As temperature rises more, even more carbon is released from the soil. This creates a cycle that keeps strengthening itself. Once this loop starts, it is difficult to stop because the system begins to add carbon to the atmosphere on its own, without needing additional human emissions. In the model, this effect becomes more important around the middle of the simulation period and makes the temperature rise faster after that point.

R4: Black Carbon Albedo Reinforcing Loop

Black carbon from sources like burning fuel and biomass travels through the air and settles on glacier surfaces. This makes the surface darker and reduces its ability to reflect sunlight. As a result, more heat is absorbed, which increases temperature and speeds up glacier melting. Studies show that black carbon plays a significant role in glacier melt, especially before the monsoon season. In the model, this effect is included as an additional contribution to melting beyond what temperature alone would cause. This loop does not directly depend on glacier size but works through the temperature and surface reflectivity pathway. Even though it acts separately, it still increases the overall rate of glacier loss.

B1: Ocean Carbon Sink Balancing Loop

As greenhouse gas levels increase, the ocean absorbs some of the carbon dioxide from the atmosphere. This helps slow down the rate at which greenhouse gases build up. However, this process becomes less effective over time. As the ocean absorbs more carbon, its capacity to take in additional carbon decreases. Because of this, the ocean can slow down the increase in greenhouse gases but cannot stop it completely. Over a long period, this balancing effect is not strong enough to counter the overall rise.

B2: Forest Carbon Sink Balancing Loop (Weakened by B3)

Forests also help remove carbon dioxide from the atmosphere by storing it in plants and soil. However, in Nepal, the amount of carbon removed by forests is very small compared to the total global emissions. This makes the effect quite limited. In addition, as glacial lakes grow and the risk of floods increases, nearby forests can be damaged or lost. Flooding puts pressure on forest areas and leads to deforestation. This reduces the already small contribution of forests in removing carbon from the atmosphere. As a result, this balancing loop becomes even weaker over time.

4 Model Formulation

4.1 Stock and Flow Diagram

Figure 3 shows the full stock and flow diagram built in Vensim. The model contains six stocks: Green House Gas, Glaciers Volume, Snow Cover, Glacial Lake, Forest Cover, and Black Carbon. Each subsystem is color coded: orange for GHG, dark red for temperature and albedo, blue for glaciers and snow, green for the glacial lake system, dark green for forest, and purple for black carbon. The simulation runs from 1980 to 2080 with a time step of 0.125 years.

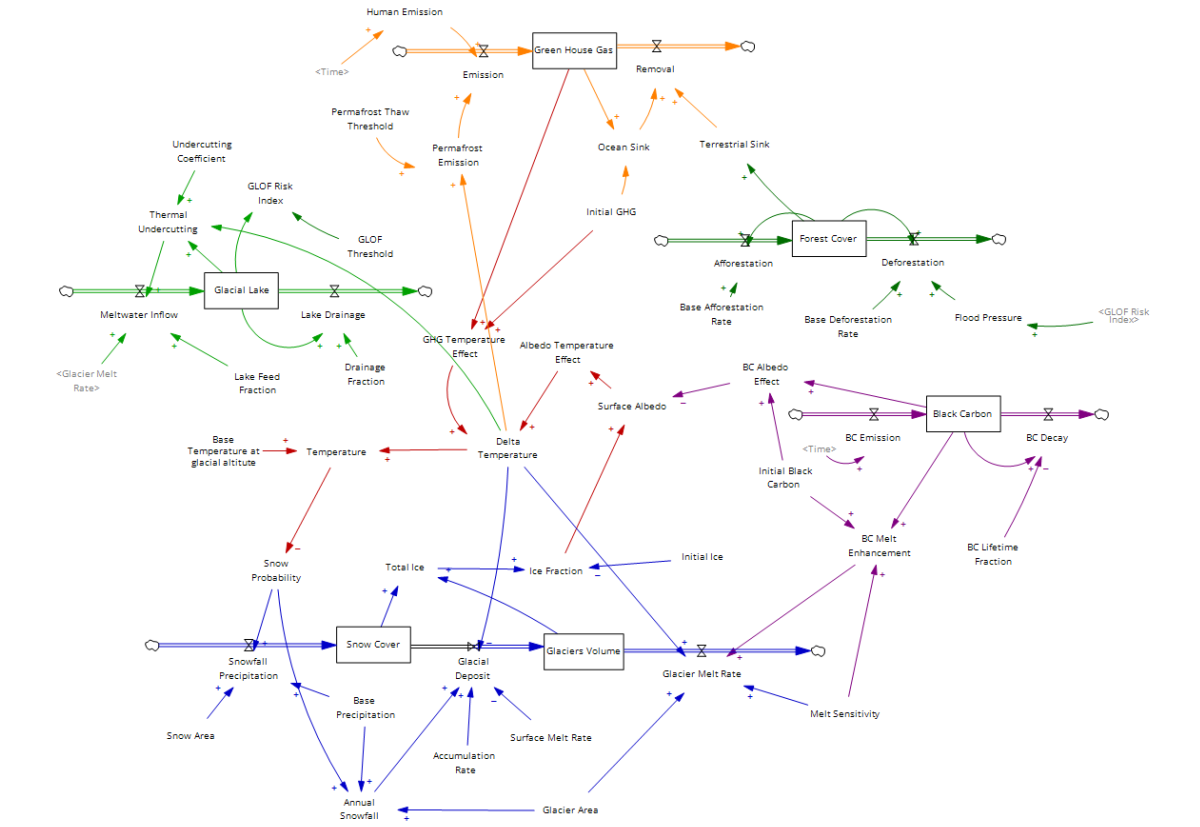

Figure 3: Stock and Flow Diagram of the Nepal glacier system

4.2 Key Equations

The three equations that are central to the model's behavior are as follows:

Equation 1: GHG Temperature Effect

$$GHGTemperatureEffect = 3 \times \frac{\ln\left(\max\left(\frac{GreenHouseGas}{InitialGHG}, 1\right)\right)}{\ln(2)}$$

This implements the logarithmic radiative forcing relationship established by Myhre et al. [2]. A climate sensitivity of 3 degrees Celsius per doubling of CO_2 is used, consistent with the likely range reported in IPCC AR6 [1]. At 1980 levels (830 Gt CO_2), the effect is zero by construction. As GHG accumulates, temperature rises in a concave fashion, slowing down relative to the linear GHG growth. The division by ln(2) correctly normalizes the sensitivity to a per-doubling basis.

Equation 2: Glacier Melt Rate

$$GlacierMeltRate = \max(0,\ (MeltSensitivity + BCMeltEnhancement) \times \Delta T \times GlacierArea)$$

This is the primary outflow from the Glaciers Volume stock. Melt Sensitivity (0.86 km per year per degree Celsius) is calibrated from field observations at Yala and Rikha Samba glaciers by Stigter et al. [7]. The BC Melt Enhancement term adds 39 percent of base sensitivity when Black Carbon is at its initial loading, consistent with ICIMOD measurements [12]. Delta Temperature is the combined warming from GHG forcing and albedo feedback. Glacier Area (0.363 km^2) converts the depth-based rate to a volumetric

flow. The MAX function prevents the flow from going negative, ensuring no spurious re-freezing at glacier scales.

Equation 3: Permafrost Emission

$$PermafrostEmission = \{ 0 \ .0154 \times (\Delta T - 1.5) if \Delta T > 1.50 otherwise$$

The coefficient 0.0154 Gt CO_2 per degree Celsius per year is derived from Schuur et al. [3], who estimate approximately 0.9 Gt CO_2 per degree of warming per year globally from permafrost thaw. We apply a Hindu Kush Himalaya fraction of roughly 1.7 percent to obtain the Nepal-relevant contribution. The threshold of 1.5 degrees Celsius represents the warming level at which permafrost thaw transitions from negligible to significant in this region.

4.3 Parameter Table

Parameter	**Value**	**Unit**	**Source**
Initial GHG	830	Gt CO_2	IPCC AR6 [1]
Initial Glaciers Volume	312	km^3	ICIMOD HI-WISE [5]
Initial Snow Cover	50	km^3	Brun et al. [8]
Initial Glacial Lake	0.5	km^3	Rounce et al. [10]
Initial Forest Cover	64,000	km^2	Global Forest Watch [15]
Initial Black Carbon	120	Gg	ICIMOD [12]
Base Temperature at glacial altitude	-15	Celsius	World Bank Nepal [17]
Melt Sensitivity	0.86	km/yr	Stigter et al. [7]
Accumulation Rate	0.55	dimensionless	Stigter et al. [7]
Permafrost Thaw Threshold	1.5	°C (ΔT)	Schuur et al. [3]
Permafrost Coefficient	0.0154	Gt CO_2/(°C yr)	Schuur et al. [3]
Ocean Sink Coefficient	10.6	Gt CO_2/yr	Friedlingstein et al. [4]
Lake Feed Fraction	0.3	dimensionless	Rounce et al. [10]
Glacier Area	0.363	km^2	ICIMOD [5]
BC Lifetime Fraction	0.5	1/yr	ICIMOD [12]
Base Afforestation Rate	0.014	1/yr	Global Forest Watch [15]
Base Deforestation Rate	0.012	1/yr	Global Forest Watch [15]
GLOF Threshold	5	km^3	ICIMOD [5]

Table 1: Key model parameters, their values, units, and sources.

4.4 Model Testing

Before running policy scenarios, we performed two basic structural tests in Vensim. First, we confirmed that both "Units" and "Model" are correct. Second, we tested boundary

behavior by setting Delta Temperature to zero for the entire simulation. In this case, Glacier Melt Rate collapses to zero and Glaciers Volume rises gradually as Glacial Deposit continues to add compacted snowfall. This is the physically expected behavior: with no warming, glaciers would slowly accumulate mass. Third, we set Human Emission to zero, confirming that GHG stock drops monotonically as ocean and terrestrial sinks continue removing pre-existing carbon. These tests give us confidence that the model's causal structure is internally consistent.

5 Policy Analysis and Results

5.1 Base Case

The base case uses all default parameter values with Human Emission growth at 1.8 percent per year (historical average from Global Carbon Project [18]) and Black Carbon emissions growing at 1.2 percent per year [12]. Figure 4 shows GHG accumulation rising from 830 Gt CO_2 in 1980 to approximately 8,395 Gt CO_2 by 2080 under the base case, a tenfold increase driven by exponential emission growth that far outpaces the logarithmically saturating ocean sink.

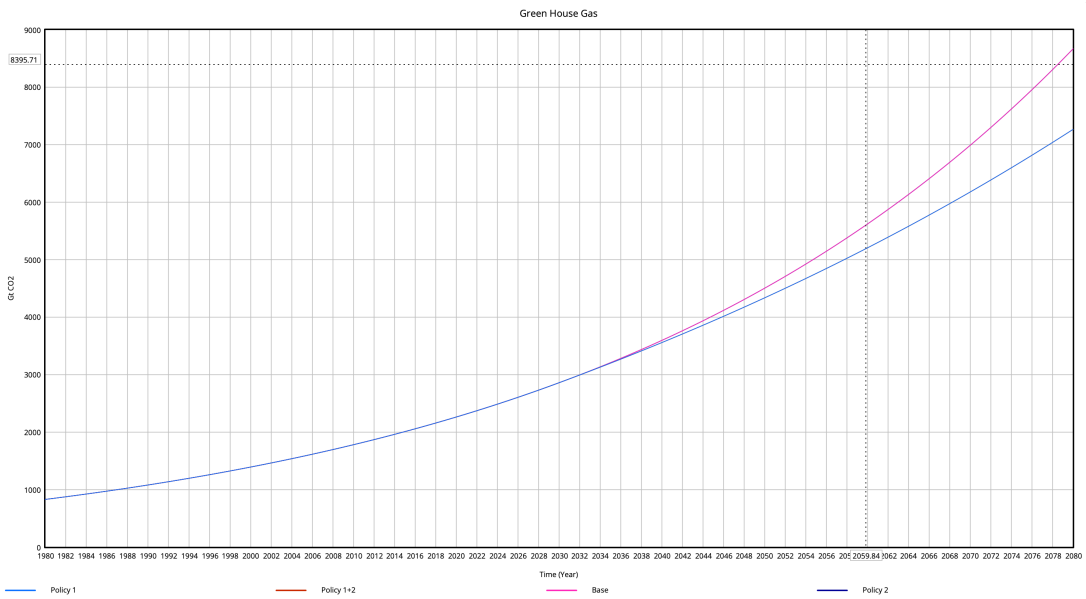

Figure 4: Green House Gas stock (Gt CO_2) from 1980 to 2080. The base case (pink) reaches approximately 8,395 Gt CO_2 by 2080. Policy 1 (light blue, emission growth reduction) substantially lowers GHG accumulation. Policy 2 (dark blue, BC reduction) and the combined scenario (red) show intermediate trajectories.

Temperature at glacier altitude rises from approximately $-14.7°C$ in 1980 to $-8.2°C$ by 2080 in the base case, representing roughly 6.5 degrees of warming over the century. As a result, Glaciers Volume falls from 312 km^3 to approximately 155 km^3, a loss of 50 percent of the 1980 baseline. The GLOF Risk Index, which starts near zero in 1980 as lake volumes are small, grows continuously and reaches 0.68 by 2080, indicating the aggregate glacial lake system is approaching two thirds of the critical volume threshold. Black Carbon grows from 120 Gg at the 1980 equilibrium to approximately 334 Gg by 2080 as South Asian emissions continue rising.

5.2 Sensitivity Analysis

To assess model robustness, we analytically evaluated the effect of a 20 percent change in two key parameters.

Melt Sensitivity (±20%): Melt Sensitivity enters the Glacier Melt Rate equation multiplicatively alongside Delta Temperature and Glacier Area. A 20 percent increase (from 0.86 to 1.032 km per year per degree Celsius) therefore increases Glacier Melt Rate proportionally at every time step. Over 100 years, this translates to a faster depletion of Glaciers Volume by an additional 15 to 20 km^3 relative to the base case. Conversely, a 20 percent reduction (0.86 to 0.688) slows melt and retains approximately 15 additional km^3 by 2080. Because Melt Sensitivity also feeds into BC Melt Enhancement (which scales proportionally to it), the BC albedo loop is correspondingly amplified or dampened, adding a secondary effect through the temperature pathway.

Human Emission initial value (±20%): A 20 percent increase in 1980 emissions (from 36 to 43.2 Gt CO_2 per year) shifts the GHG trajectory upward from the first year, reaching a higher plateau by 2080 and bringing forward the crossing of the 1.5 degree permafrost threshold by several years. This activates the R3 permafrost loop earlier, causing a steeper acceleration of warming in the second half of the simulation. A 20 percent reduction delays the permafrost threshold crossing and reduces the terminal GHG stock by roughly 800 Gt CO_2 by 2080. The sensitivity to this parameter confirms that early action on emission reductions delivers compounding benefits through the permafrost feedback pathway.

5.3 Policy 1: Human Emission Growth Rate Reduction

This policy reduces the emission growth rate from 1.8 percent per year to 1.0 percent per year starting in 2030, representing a moderate decarbonization trajectory consistent with the IEA Stated Policies Scenario [19]. Emissions still grow, but more slowly. The effect on GHG accumulation is substantial: by 2080, GHG reaches approximately 7,300 Gt CO_2 rather than 8,395 Gt CO_2, a difference of roughly 1,100 Gt CO_2. This moderates the GHG Temperature Effect and slows warming. However, the effect on Glaciers Volume is limited, as shown in Figure 5. The lines for Policy 1 and the base case track closely through 2060, diverging only in the final two decades. By 2080, Policy 1 retains approximately 163 km^3 compared to 155 km^3 in the base case, a modest improvement of 8 km^3.

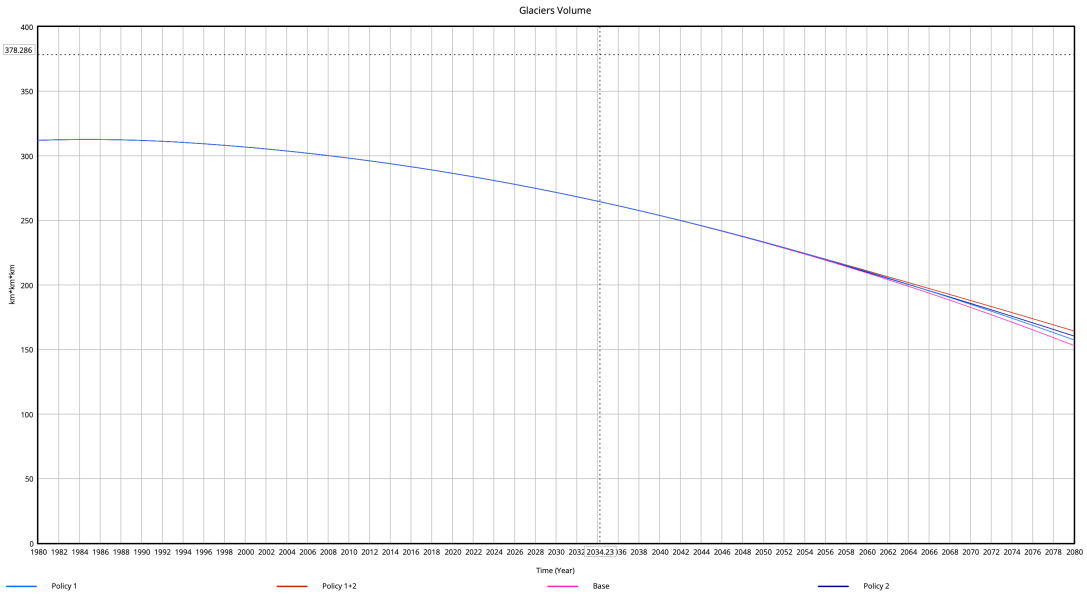

Figure 5: Glaciers Volume (km^3) comparing base case (pink), Policy 1 (light blue, emission growth reduction), Policy 2 (dark blue, BC reduction), and combined Policy 1 plus 2 (red). Policies diverge from the base only after 2060, highlighting the inertia of the climate system.

This limited glacier response despite a substantial GHG reduction reflects the thermal inertia of the climate system. Warming already accumulated before 2030 continues driving melt, and the temperature reduction from slower GHG growth only becomes meaningful toward the end of the century. Nevertheless, this policy delivers a long term benefit beyond 2080 as the lower GHG trajectory reduces the probability of crossing the permafrost tipping point.

5.4 Policy 2: Black Carbon Emission Growth Rate Reversal

This policy reverses BC emission growth from plus 1.2 percent per year to minus 3 percent per year after 2030, representing aggressive deployment of clean brick kiln technology, diesel emission standards, and elimination of crop residue burning across the Indo-Gangetic Plain. ICIMOD research suggests that a 30 to 50 percent reduction in BC is achievable within 15 years through existing technology in South Asia [12], and Bond et al. [13] confirm the technical feasibility of this scale of mitigation. Because BC has an atmospheric lifetime of approximately two years (BC Lifetime Fraction = 0.5 per year), the BC stock responds quickly to changes in emission rates.

Figure 6 shows that Black Carbon in the base case and Policy 1 scenarios both grow continuously (overlapping light blue and pink lines), reaching over 300 Gg by 2080. Under Policy 2 and the combined scenario (Policy 1 plus 2), Black Carbon peaks around 2030 at approximately 215 Gg and then falls sharply to approximately 55 Gg by 2080.

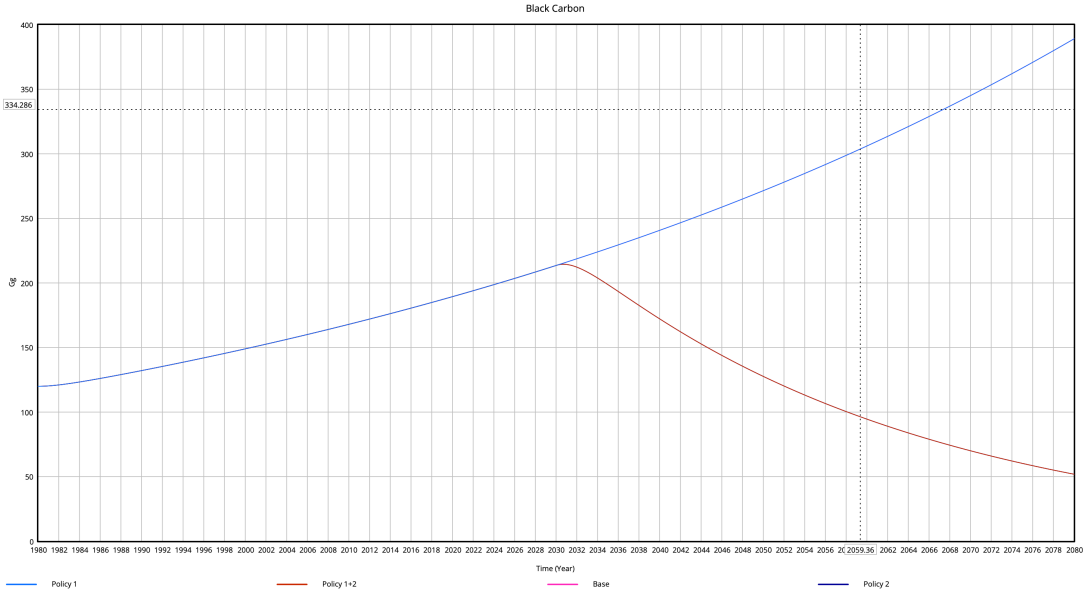

Figure 6: Black Carbon stock (Gg) under all scenarios. Policy 2 (and Policy 1 plus 2, shown in red) reverses BC growth after 2030, causing the stock to fall from a peak of approximately 215 Gg to around 55 Gg by 2080. The base case and Policy 1 (which does not target BC) overlap on the rising trajectory.

The most pronounced benefit of BC reduction is on the GLOF Risk Index. Figure 7 shows that the base case reaches a GLOF Risk Index of 0.68 by 2080, while Policy 2 and the combined scenario reduce this to approximately 0.57. The mechanism is: lower BC reduces BC Albedo Effect, raising Surface Albedo, which reduces Albedo Temperature Effect and therefore Delta Temperature. A cooler Delta Temperature slows Glacier Melt Rate, which reduces Meltwater Inflow to the glacial lake, which slows lake growth and limits the activation of the Thermal Undercutting reinforcing loop. This is a particularly important result because GLOF events pose direct threats to infrastructure, agriculture, and human lives in Nepal's river valleys [16].

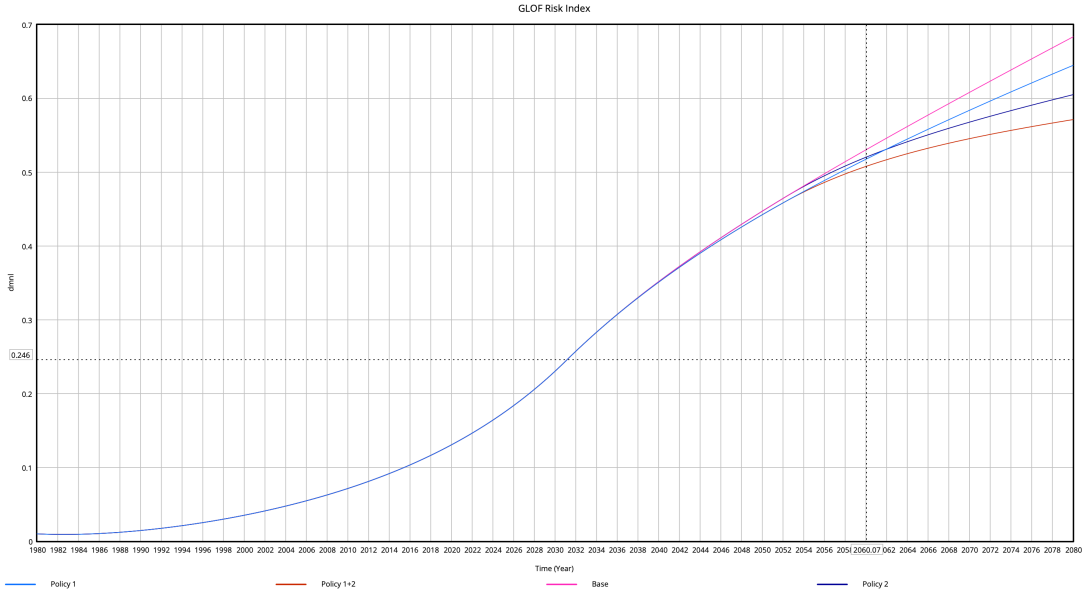

Figure 7: GLOF Risk Index (dimensionless, 0 to 1) from 1980 to 2080. The base case (pink) reaches 0.68 by 2080. Policy 2 and the combined scenario (red) reduce this to approximately 0.57. The index follows an accelerating growth path driven by the GLOF undercutting reinforcing loop (R2).

5.5 Bonus Exploration: Melt Sensitivity Reduction (Policy 3)

As an exploratory extension, we examined a physical glacier protection intervention in which Melt Sensitivity is reduced from 0.86 to 0.602 km per year per degree Celsius after 2030, representing a 30 percent reduction. This simulates the partial coverage of glacier surfaces with reflective geotextile blankets, an approach that has demonstrated 50 to 70 percent melt reduction on covered areas in field experiments [20, 21]. The 30 percent system-level figure is conservative, reflecting the reality that only a fraction of Nepal's 3,902 km^2 glacier area can realistically be covered.

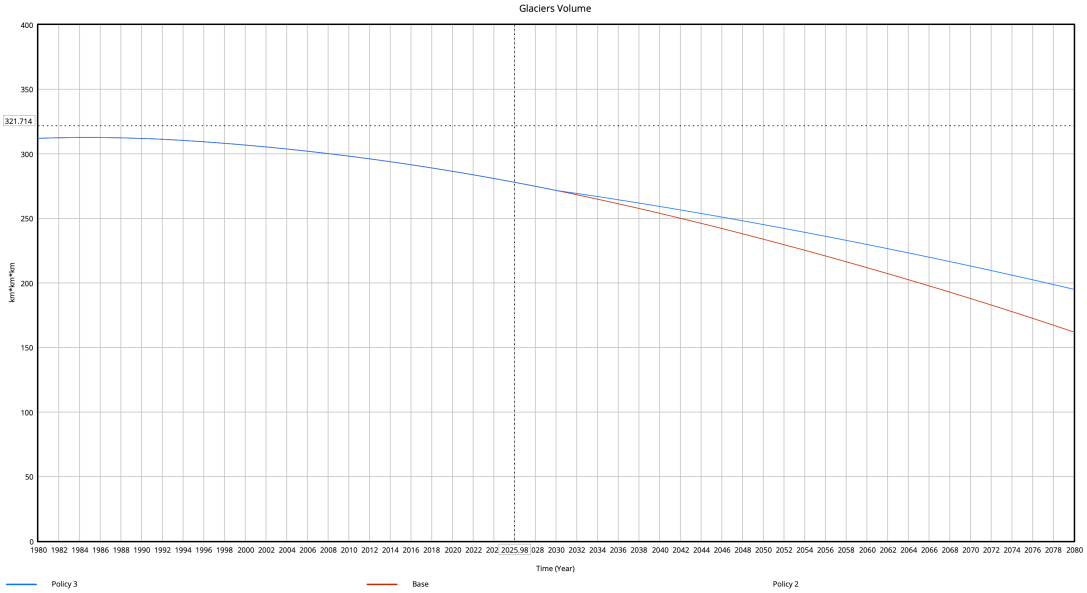

Figure 8: Glaciers Volume under Policy 3 (Melt Sensitivity reduction, shown in blue) versus base case (red). Policy 3 retains approximately 197 km^3 by 2080, compared to 165 km^3 in the base case, preserving an additional 32 km^3 of glacier ice.

As Figure 8 shows, this intervention retains approximately 197 km^3 of glacier ice by 2080, compared to 165 km^3 in the base case, a preservation of an additional 32 km^3 or roughly 19 percent more ice. This is a larger glacier volume improvement than either Policy 1 or Policy 2 achieve on their own. However, this policy was considered a bonus exploration rather than a primary recommendation because large scale physical glacier protection is extremely costly, logistically challenging across 3,902 km^2 of high altitude terrain, and not scalable as a primary climate response strategy.

6 Conclusions and Recommendations

6.1 What Happens in the Base Case

If current trends continue, the glacier system in Nepal shows strong and worsening decline over time. Glacier volume drops from 312 km^3 to about 155 km^3, which is nearly half of the original ice. At the same time, greenhouse gases increase sharply from 830 to more than 8,000 Gt CO_2. The GLOF Risk Index rises to around 0.68, meaning the glacial lakes reach a level close to dangerous limits. This creates serious risks for people, farms, and infrastructure in areas below the glaciers. Temperature in glacier regions rises by about 6.5 degrees Celsius over the century. This increase is high enough to trigger permafrost thaw, which then adds more carbon to the atmosphere and makes warming even worse in the later part of the timeline.

6.2 Which Policy Worked Best

To reduce the risk of glacial lake floods, which is the most immediate concern, reducing black carbon emissions works best. This policy lowers the GLOF Risk Index from 0.68 to 0.57 by 2080 and slows the growth of glacial lakes by weakening the albedo related feedback. On the other hand, reducing the growth of emissions is more effective for long

term control of greenhouse gases. It reduces total GHG levels by more than 1,100 Gt CO_2 by 2080. When both policies are applied together, the results are better across all parts of the system. So the best approach is to combine both actions. One reduces the temperature increase, and the other reduces how strongly glaciers respond to that temperature.

6.3 Side Effects and Unintended Consequences

One side effect appears in the forest system. As the risk of glacial lake floods increases, nearby forests face more pressure from flooding. This leads to deforestation, which reduces forest cover. Since forests absorb carbon, this reduces their ability to remove carbon from the atmosphere. This creates another chain where more melting increases flood risk, which leads to forest loss, which slightly increases greenhouse gases. Although this effect is small compared to global emissions, it shows that reducing flood risk also helps protect forests. This benefit is often ignored in climate policy discussions.

Another observation is that reducing black carbon greatly lowers pollution levels but has only a limited effect on preserving glacier volume. This is because black carbon mainly affects melting through surface darkening, while temperature plays a larger role in overall glacier loss. So black carbon reduction should be seen more as a way to reduce flood risk and protect water systems rather than as a direct way to save glaciers.

6.4 Learning and Inferences

An important lesson from this model is that climate systems respond slowly. Policies introduced around 2030 do not show clear results until much later, often after 2060. This delay can make policies seem ineffective in the short term even though they are important in the long run. Another key point is that reinforcing feedback loops, especially those related to albedo and glacial lakes, play a major role in speeding up glacier loss. If these loops are weakened, the overall system benefits. Reducing black carbon is one such action because it weakens multiple feedback effects and gives good results compared to its cost.

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A Full Equation List (Exported from Vensim Model)

```
Green House Gas = INTEG(Emission - Removal, 830)
    ~ Gt CO2

Emission = Human Emission + Permafrost Emission
    ~ Gt CO2/Year

Removal = Ocean Sink + Terrestrial Sink
    ~ Gt CO2/Year

Human Emission = 36 * EXP(Emission Growth Rate * (Time - 1980))
    ~ Gt CO2/Year

Emission Growth Rate = 0.018
    ~ 1/Year

Permafrost Emission = IF THEN ELSE(Delta Temperature > Permafrost Thaw Threshold,
    0.037 * (Delta Temperature - Permafrost Thaw Threshold), 0)
    ~ Gt CO2/Year

Ocean Sink = 10.6 * LN(Green House Gas / Initial GHG + 1)
    ~ Gt CO2/Year

Terrestrial Sink = 3.1 + 3.9e-08 * Forest Cover
    ~ Gt CO2/Year

Permafrost Thaw Threshold = 1.5
    ~ dmnl

Initial GHG = 830
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    ~ Gt CO2

Temperature = Base Temperature at glacial altitute + Delta Temperature
    ~ Celsius

Base Temperature at glacial altitute = -15
    ~ Celsius

Delta Temperature = GHG Temperature Effect + Albedo Temperature Effect
    ~ dmnl

GHG Temperature Effect = 3 * LN(MAX(Green House Gas / Initial GHG, 1)) / LN(2)
    ~ dmnl

Albedo Temperature Effect = 1.5 * (0.62 - Surface Albedo)
    ~ dmnl

Surface Albedo = MAX(0.15, 0.25 + 0.37 * Ice Fraction - BC Albedo Effect)
    ~ dmnl

BC Albedo Effect = 0.038 * MIN(1, Black Carbon / Initial Black Carbon)
    ~ dmnl

Ice Fraction = MIN(1, MAX(0, Total Ice / Initial Ice))
    ~ dmnl

Total Ice = Glaciers Volume + Snow Cover
    ~ km*km*km

Initial Ice = 362
    ~ km*km*km

Glaciers Volume = INTEG(Glacial Deposit - Glacier Melt Rate, 312)
    ~ km*km*km

Snow Cover = INTEG(Snowfall Precipitation - Glacial Deposit, 50)
    ~ km*km*km

Glacial Deposit = MAX(0, Accumulation Rate * Annual Snowfall
    - Surface Melt Rate * Delta Temperature)
    ~ km*km*km/Year

Annual Snowfall = Snow Probability * Base Precipitation * Glacier Area
    ~ km*km*km/Year

Glacier Melt Rate = MAX(0, (Melt Sensitivity + BC Melt Enhancement)
    * Delta Temperature * Glacier Area)
    ~ km*km*km/Year
```

```
Melt Sensitivity = 0.86
    ~ km/(Year*Celsius)

BC Melt Enhancement = 0.39 * Melt Sensitivity * MIN(1, Black Carbon / Initial Black C
    ~ km/(Year*Celsius)

Snowfall Precipitation = Snow Probability * Base Precipitation * Snow Area
    ~ km*km*km/Year

Snow Probability = 1 / (1 + EXP(1.2 * (Temperature - 3)))
    ~ dmnl

Accumulation Rate = 0.55
    ~ dmnl

Surface Melt Rate = 0.25
    ~ km*km*km/(Year*Celsius)

Glacier Area = 0.363
    ~ km*km

Base Precipitation = 1.5
    ~ km/Year

Snow Area = 0.8
    ~ km*km

Glacial Lake = INTEG(Meltwater Inflow - Lake Drainage, 0.5)
    ~ km*km*km

Meltwater Inflow = Lake Feed Fraction * Glacier Melt Rate + Thermal Undercutting
    ~ km*km*km/Year

Lake Drainage = Drainage Fraction * Glacial Lake
    ~ km*km*km/Year

Thermal Undercutting = Undercutting Coefficient * Glacial Lake * MAX(0, Delta Tempera
    ~ km*km*km/Year

GLOF Risk Index = MIN(1, Glacial Lake / GLOF Threshold)
    ~ dmnl

GLOF Threshold = 5
    ~ km*km*km

Lake Feed Fraction = 0.3
    ~ dmnl
```

```
Drainage Fraction = 0.08
    ~ 1/Year

Undercutting Coefficient = 0.025
    ~ 1/Year

Forest Cover = INTEG(Afforestation - Deforestation, 64000)
    ~ km*km

Afforestation = Base Afforestation Rate * Forest Cover
    ~ km*km/Year

Deforestation = (Base Deforestation Rate + Flood Pressure) * Forest Cover
    ~ km*km/Year

Base Afforestation Rate = 0.014
    ~ 1/Year

Base Deforestation Rate = 0.012
    ~ 1/Year

Flood Pressure = 0.002 * GLOF Risk Index
    ~ 1/Year

Black Carbon = INTEG(BC Emission - BC Decay, 120)
    ~ Gg

BC Emission = 60 * EXP(BC Growth Rate * (Time - 1980))
    ~ Gg/Year

BC Growth Rate = 0.012
    ~ 1/Year

BC Decay = BC Lifetime Fraction * Black Carbon
    ~ Gg/Year

BC Lifetime Fraction = 0.5
    ~ 1/Year

Initial Black Carbon = 120
    ~ Gg

FINAL TIME = 2080
INITIAL TIME = 1980
TIME STEP = 0.125
SAVEPER = TIME STEP
```

B Past Reports

Intermediate Project Submission: System Dynamics Modeling of the Effect of Greenhouse Gases on Ice Cover

Introduction and What We Updated

This report covers the recent updates we made to our System Dynamics Model, which looks at how greenhouse gases are causing global ice cover and glaciers to melt. The main goal of this update was to take our project from a basic visual concept and turn it into a working mathematical simulation.

In our first draft, we mainly focused on drawing out the ideas to show how things like melting ice and global temperatures are connected. Now, we have actually built the math behind it so we can test different scenarios. We added three major updates to make the model more realistic: we changed how we calculate the warming effect of carbon dioxide so it matches real physics, we broke down how the Earth stores carbon into different categories, and we added a tipping point for melting permafrost. Together, these changes give us a much better picture of how fast global ice is melting and how bad the resulting environmental risks, like rising sea levels and flooding, could get.

Details of Our Model Changes

Moving from Diagrams to Math Equations

We took our original Causal Loop Diagrams (which just had arrows showing what affects what) and converted them into actual Stock and Flow models. Instead of just saying more carbon means more heat, we are now using differential equations to calculate exactly how much carbon is building up and how fast the ice is depleting over time. We also tweaked our starting numbers to match real global mass-balance data, so our model's predictions make sense on a worldwide scale.

Calculating Temperature Changes Logarithmically

Before, we basically assumed that adding more CO_2 to the atmosphere would increase the temperature in a straight, steady line. But in reality, the warming effect of CO_2 actually slows down a little bit as the atmosphere gets more crowded with it. We updated our formulas to calculate the temperature change logarithmically. Think of it like adding blankets when you are cold: the first few blankets make a huge difference, but adding a tenth blanket does not warm you up as much as the first one did. This makes our model match up with real atmospheric science.

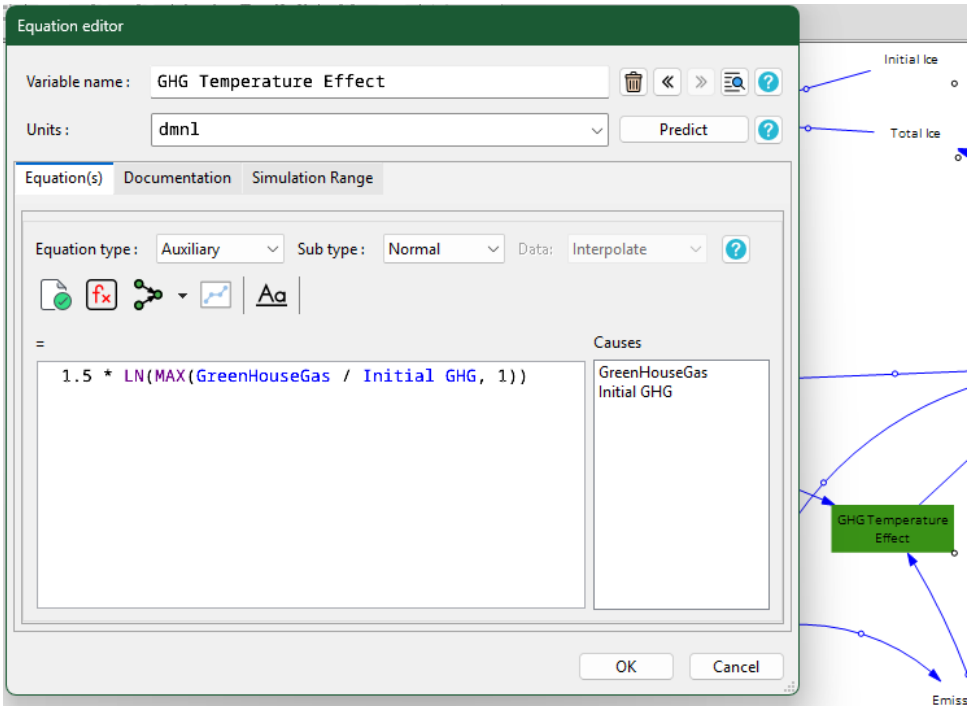

Figure 9: A screenshot of our Stock and Flow model showing the logarithmic math for climate forcing. The GHG Temperature Effect variable uses the Myhre et al. (1998) logarithmic relationship to translate CO_2 concentration into degrees of warming.

Making the Carbon Cycle More Realistic

In our first model, we treated afforestation like a fix that could absorb unlimited amounts of carbon. We realized that was not realistic. So, we updated the model to split the Earth's carbon sinks into two real parts: the ocean and the land. We added limits to how much these sinks can hold. For example, as the ocean absorbs more carbon, it becomes too acidic to keep absorbing it at the same rate. This helps us see that while planting trees is valuable, nature has limits on how much of our pollution it can clean up over the next 50 to 100 years.

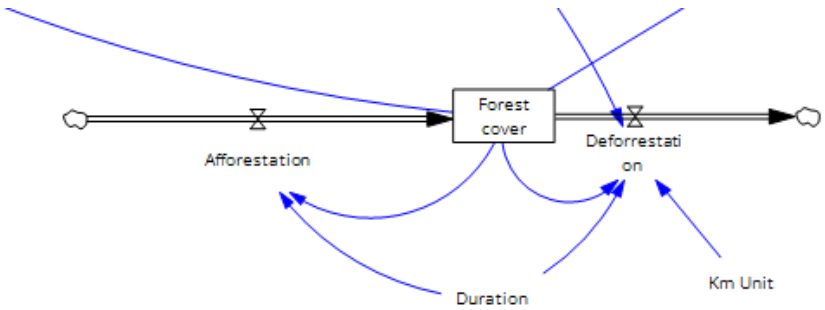

Figure 10: Our updated diagram showing the different carbon sinks (ocean and land) and their saturation limits within the GHG subsystem of the Stock and Flow model.

Adding the Permafrost Melting Trap

We added a critical new feedback loop: the permafrost thaw tipping point. Permafrost is frozen ground that has large quantities of old greenhouse gases trapped inside it. In our

new model, once the global temperature warming anomaly exceeds 1.5 degrees Celsius above the 1980 baseline, this frozen ground starts to melt and release trapped carbon dioxide.

This creates a dangerous trap. Once the permafrost starts melting, the Earth begins warming itself up, even if humans stop polluting. This causes a significant increase in our simulation's GHG concentrations, which leads to faster glacier melt and growing flood risks later in the simulation.

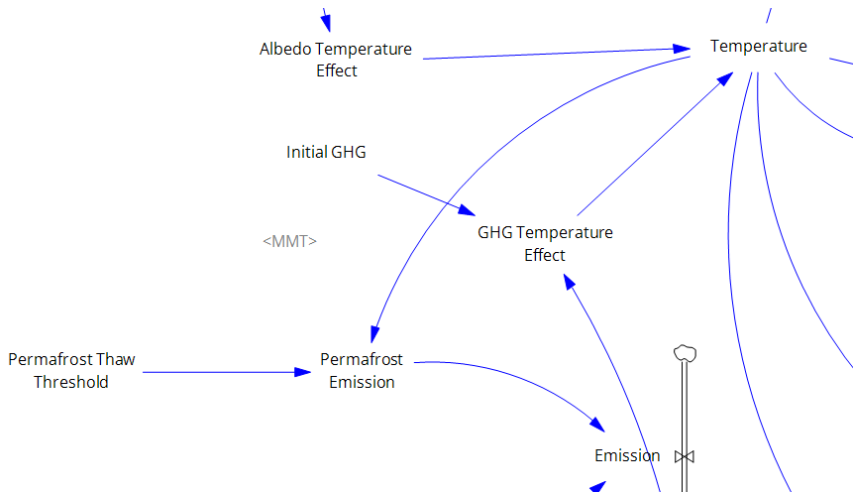

Figure 11: The section of our model that simulates permafrost melting and releasing additional CO_2 once Delta Temperature crosses the 1.5 degree Celsius threshold defined by Schuur et al. (2015).

Key Model Variables, Stocks, and Units

To make sure our simulation works mathematically, we had to assign specific units and roles to every part of the system. The table below summarizes the main variables, stocks, and flows in our updated model.

Variable Name	Unit	What it Does in the Model
Stocks (Things that Accumulate)		
Atmospheric Carbon	Gt CO_2	The total amount of carbon trapped in the atmosphere, which drives the warming effect.
Glaciers Volume	km^3	The total remaining volume of Nepal's glacier ice.
Snow Cover	km^3	The seasonal snowpack that compacts into glacier ice over time.
Glacial Lake	km^3	The aggregate volume of proglacial lakes, driving GLOF risk.
Forest Cover	km^2	Total forested area in Nepal, which provides a small terrestrial carbon sink.
Black Carbon	Gg	Atmospheric black carbon stock depositing on glacier surfaces and reducing albedo.
Flows (Rates of Change over Time)		
Human Emission	Gt CO_2/yr	The rate at which human activity adds new carbon to the atmosphere every year.
Glacier Melt Rate	km^3/yr	How fast the glacier ice volume is melting away due to warming and black carbon.
Glacial Deposit	km^3/yr	How fast compacted snowfall transfers from Snow Cover into Glaciers Volume.
Ocean Sink	Gt CO_2/yr	The rate at which the ocean absorbs carbon from the atmosphere.
Auxiliary Variables (Calculations and Triggers)		
GHG Temperature Effect	°C	The warming contribution from CO_2 radiative forcing, calculated logarithmically.
Delta Temperature	°C	Total warming anomaly above the 1980 baseline, combining GHG and albedo effects.
Surface Albedo	dimensionless	A fraction showing how reflective the glacier surface is. It drops as ice melts and black carbon deposits.
GLOF Risk Index	dimensionless	A normalized index from 0 to 1 representing glacial lake outburst flood risk.

Table 2: A summary of the main components driving the Nepal glacier System Dynamics simulation.

References for the Intermediate Submission

1. Myhre, G., Highwood, E.J., Shine, K.P., and Stordal, F. (1998). New estimates of radiative forcing due to well mixed greenhouse gases. *Geophysical Research Letters*, 25(14), 2715–2718.

2. Schuur, E.A.G., McGuire, A.D., Schadel, C., et al. (2015). Climate change and the

permafrost carbon feedback. *Nature*, 520(7546), 171–179.

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4. IPCC (2021). *Climate Change 2021: The Physical Science Basis*. Cambridge University Press.

5. Brun, F., Berthier, E., Wagnon, P., Kaab, A., and Treichler, D. (2017). A spatially resolved estimate of High Mountain Asia glacier mass balances from 2000 to 2016. *Nature Geoscience*, 10(9), 668–673.

6. Bolch, T., et al. (2012). The State and Fate of Himalayan Glaciers. *Science*, 336(6079), 310–314.

Original Preliminary Report

Project Summary

This project investigates the direct effect of greenhouse gases on global ice cover using a system dynamics framework. By constructing a causal model, the research simulates how anthropogenic carbon emissions drive atmospheric warming, which in turn accelerates ice mass degradation worldwide. The simulation captures crucial hydrological transitions, tracking the conversion of stored glacial and polar ice into surface runoff and evaluating the subsequent risks of altered streamflows, rising sea levels, and downstream flooding.

A central feature of the model is the integration of physical feedback mechanisms, specifically the albedo effect. As increasing greenhouse gas concentrations reduce overall ice cover, the exposed darker surfaces absorb more solar radiation, amplifying global warming. Furthermore, the project incorporates biological variables, such as global afforestation and carbon sequestration rates, to evaluate their potential in mitigating these warming trends. By running various greenhouse gas emission scenarios, the simulation aims to pinpoint irreversible tipping points in global ice melt and assess the effectiveness of systemic environmental interventions.

Introduction

Glaciers and polar ice caps are highly sensitive barometers of global climate health. Over the past century, the exponential rise in anthropogenic greenhouse gas emissions has fundamentally altered the Earth's energy balance. Gases such as carbon dioxide and methane trap outgoing infrared radiation, leading to a steady increase in mean global temperatures. This thermal forcing has a profound, accelerated effect on cryospheres worldwide.

The Earth's polar regions and high-altitude mountain ranges harbor massive volumes of terrestrial ice, serving as critical regulators of the Earth's climate system. These ice reserves act as massive frozen reservoirs that regulate global sea levels and regional hydrology, sustaining major river networks worldwide. However, elevated concentrations of greenhouse gases, compounded by local pollutants like black carbon in mountainous regions, are driving rapid glacial retreat. While the initial phase of this melting can temporarily swell river volumes, the ultimate depletion of these ice reserves threatens long-term water security for millions and risks catastrophic coastal flooding. Understanding the complex, nonlinear relationship between continuous GHG emissions and global ice stability requires an integrated modeling approach, which this project addresses through system dynamics.

Literature Survey and Research Gap

The accelerated degradation of high-altitude and polar ice due to anthropogenic forcing is well-documented in contemporary climate science. The Intergovernmental Panel on Climate Change (IPCC, 2021) emphasizes that human-induced greenhouse gas emissions are the primary driver of the pervasive retreat of mountain glaciers and ice sheets globally. Studies utilizing satellite imagery provide spatially resolved estimates of glacier mass balance, confirming significant and widespread mass loss across vulnerable regions like the Himalayas (Brun et al., 2017) and polar caps.

Similarly, regional studies highlight that while local climatic variations exist, the overarching fate of global glaciers is dominated by sustained mass loss driven by rising tem-

peratures (Bolch et al., 2012). This warming trajectory not only threatens freshwater availability but also alters the global hazard landscape. For instance, recent reports indicate a significant expansion in glacial lake areas in mountainous regions, greatly amplifying the risk of Glacial Lake Outburst Floods [16]. Furthermore, projections suggest that a 3 degree Celsius rise in global temperatures could subject vast regions to extended drought conditions, simultaneously disrupting seasonal snow accumulation and agricultural stability worldwide.

Research Gap: Despite extensive observational data confirming the adverse effect of greenhouse gases on ice cover, much of the literature remains siloed. Hydrological studies, climate forcing models, and socio-economic impact assessments are rarely integrated. Specifically, there is a deficit in predictive models that actively link continuous GHG emission pathways with real-time feedback loops like the albedo effect and ecological interventions within a single framework. This project fills that gap by deploying a comprehensive system dynamics model that synthesizes climate drivers, total ice mass balance, and potential mitigation strategies.

Project Objectives

- To develop an integrated system dynamics model that explicitly maps the causal relationship between greenhouse gas concentrations and global ice melt rates.
- To quantify the amplifying role of the ice-albedo feedback loop under varying anthropogenic emission scenarios.
- To simulate the downstream hydrological consequences, specifically tracking transitions from temporary runoff surges to eventual long-term water scarcity.
- To evaluate the mitigating potential of systemic interventions, such as carbon sequestration via afforestation, on stabilizing ice environments.

Proposed Methodology

The research relies on System Dynamics modeling in Vensim to capture the complex, interdependent variables affecting global ice ecosystems.

Causal Loop Modeling: We mapped out both balancing and reinforcing feedback mechanisms. The core reinforcing loop demonstrates how GHG-induced temperature increases cause ice melt, which lowers surface albedo, thereby increasing thermal absorption and further accelerating melt.

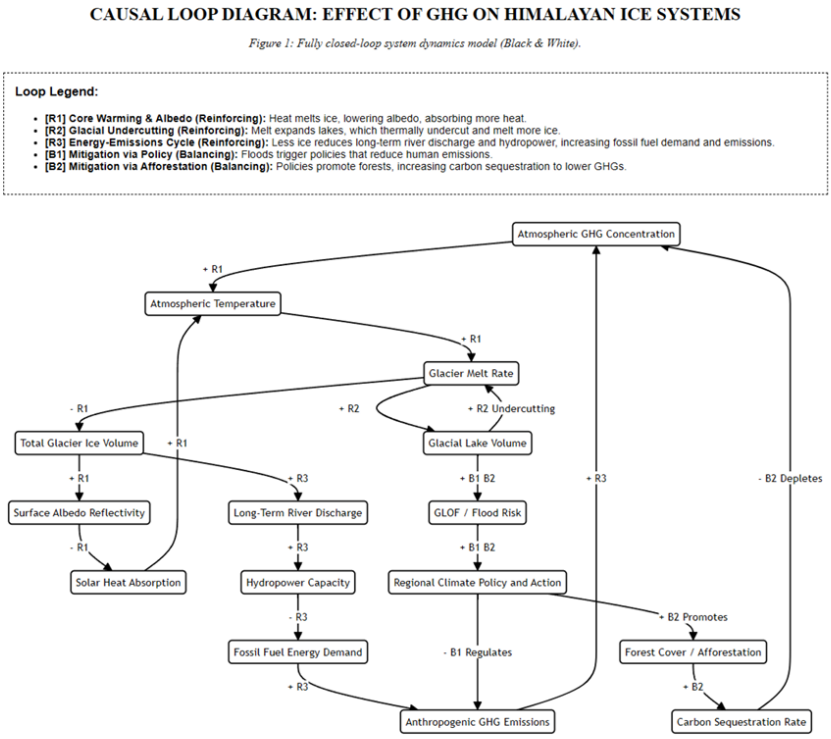

Figure 12: Causal Loop Diagram from the preliminary report, illustrating the feedback mechanisms between emissions, warming, and global ice degradation.

Stock and Flow Structures: The model includes Atmospheric GHG Load, Glacier and Snow Cover volumes, Glacial Lake, Forest Cover, and Black Carbon as stocks. Flows capture the rate of carbon emissions, snowfall accumulation, glacier ablation, meltwater inflow, lake drainage, afforestation, deforestation, and black carbon emission and decay.

Expected Outcomes

- Identification of critical ecological tipping points where the rate of ice melting permanently outpaces winter accumulation due to sustained GHG levels.
- A robust visual and mathematical demonstration of the peak water phenomenon, the point at which increased river flow from accelerated melting reverses into chronic water shortages.
- Data-driven insights into how aggressively emission reductions must be implemented to preserve the remaining Himalayan ice systems.

References for the Preliminary Report

1. IPCC (2021). *Climate Change 2021: The Physical Science Basis*. Cambridge University Press.
2. Brun, F., Berthier, E., Wagnon, P., Kaab, A., and Treichler, D. (2017). A spatially resolved estimate of High Mountain Asia glacier mass balances from 2000 to 2016. *Nature Geoscience*, 10(9), 668–673.

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6. The Hindu (2023). Himalayan glaciers could lose 80 percent of their volume if global warming is not controlled.

7. The Hindu (2023). Rising temperatures are leading to torrential rains in the Himalayas.

C Contribution Matrix

Project Task	**Deliverables**	**A (%)**	**B (%)**	**C (%)**	**D (%)**	**Total**
Conceptualization and Structural Mapping	Boundary definition, Reference Modes, Dynamic Hypothesis, CLDs, Feedback Loop identification					100%
Model Formulation, Testing, and Policy Analysis	SFD equations in Vensim, Unit checking, debugging, Base case, Scenario design, Sensitivity Analysis					100%
Report and Presentation	Final report writing, visualization, presentation and formatting					100%

Table 3: Contribution matrix. Team members: A = Anishek Kumar Chaudhary, B = Pratik Man Shah, C = Ritika Jaiswal, D = Rohit Agrawal.

Team Sign-off:

Student A: _____ Student B: _____

Student C: _____ Student D: _____